

Date:	15 March 2026	Team ID:	SWTID-2026-6684
Project Name:	ComicCraft - AI Comic Story Creator using Gemini Models		
Maximum Marks:	5 Marks		

Functional Features Overview: This document lists all the functional features included in the **ComicCraft** project solution. Each feature is described with its purpose, implementation status, module mapping, and contribution to the overall solution.

Functional Features List

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	Story Configuration Form	Captures user prompt, main character, setting, story tone, and visual art style parameters.	Frontend UI / index.html	Done	0.6
2	5-Panel JSON Outline Engine	Queries Gemini 1.5 Flash to generate structured panel titles, descriptions, and image prompts.	AI Pipeline / gemini_flash.py	Done	0.7
3	Narration & Dialogue Generator	Employs Gemini 1.5 Pro to write creative narrative captions and character dialogues for each panel.	AI Pipeline / gemini_pro.py	Done	0.7
4	Panel Illustration Renderer	Generates high-quality comic panel artwork matching descriptions using Stable Diffusion.	AI Pipeline / image_generator.py	Done	0.7
5	Interactive Comic Preview	Renders complete panel sequence with generated artwork, narration, and dialogue bubbles.	Frontend UI / comic_preview.html	Done	0.6
6	PDF Export & Formatting	Compiles comic images and story text into a downloadable PDF document using FPDF.	Export Engine / exporters.py	Done	0.6
7	REST API Endpoint	Provides programmatic POST endpoint (/generate-comic/json) for JSON clients.	Backend Router / routes.py	Done	0.6
8	Export Confirmation & Download	Provides a clean confirmation screen with local download triggers for generated PDFs.	Frontend UI / export_success.html	Done	0.5

Feature Implementation Summary

8
Total Features Planned

8
Total Features Implemented

6
Core / Must-Have Features

2

Additional / Nice-to-Have

8 / 8 (100%)

Features Tested & Verified

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Story Input Form (index.html), Comic Preview (comic_preview.html), Export Success (export_success.html)
2	Backend / Logic	2	FastAPI route handling (routes.py), PDF compilation via FPDF (exporters.py)
3	Database / Storage	1	Local file system storage for rendered panels (static/panels/) and PDFs (static/exports/)
4	API / Integration	2	Gemini 1.5 Flash JSON outline API, Gemini 1.5 Pro story text API, Diffusers Stable Diffusion API
5	Security / Authentication	1	Environment variable security via .env file for API keys (`GEMINI_API_KEY`, `HF_API_KEY`)

Evaluation Verification

All planned functional features for project **ComicCraft - AI Comic Story Creator using Gemini Models** (Team ID: SWTID-2026-6684) have been fully implemented, verified through automated integration testing, and documented according to SmartBridge specifications.